



The
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SPD Protect Action



- **SPD** stands for **Security Policy Database**, and **SA** stands for **Security Association** — which is like a set of rules for how to secure a packet (e.g., encrypt or authenticate it).
- Now, here's what happens when a packet needs to be protected, but the **SA is not available**:

Outbound Processing (Sending a Packet):

- The system **tries to send a packet**.
- But it checks and sees that there's **no Security Association (SA)** for it yet.
- So, it will **automatically use IKE** (Internet Key Exchange) to **dynamically create the SA**.
- Once the SA is created, the packet can be sent securely.

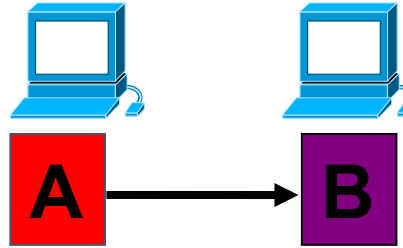
In short: If you're sending a packet and no SA exists, the system will try to **create one automatically using IKE**.

- A packet is **received** that needs to be protected (based on the policy).
- The system checks for a matching **SA**.
- If there is **no matching SA**, the system **drops (discards)** the packet — it doesn't process or accept it.

In short: If a protected packet comes in and there's no SA, the system **throws it away** to stay secure.

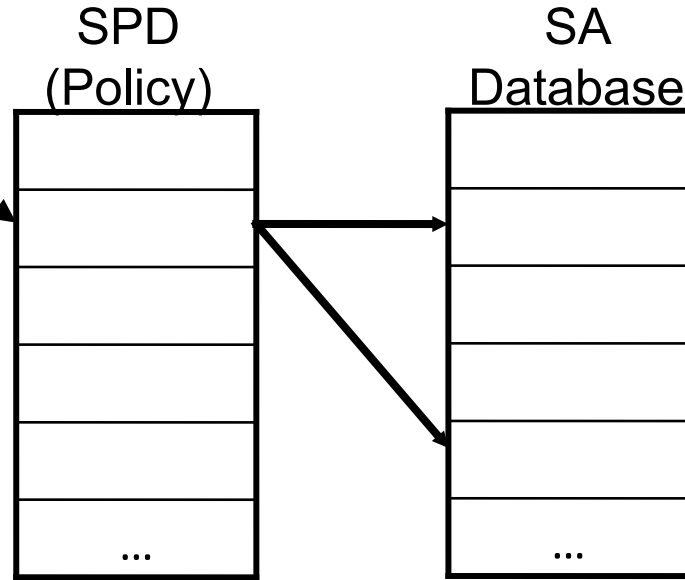
Outbound Processing

Outbound packet (on A)



IP Packet

*Is it for IPSec?
If so, which
policy
entry to select?*



*Determine the SA
and its SPI*

*IPSec
processing*

**SPI &
IPSec
Packet**



Send to B

Check Policy (SPD) – The system checks if the IP packet needs IPsec (encryption/authentication) based on the Security Policy Database (SPD).

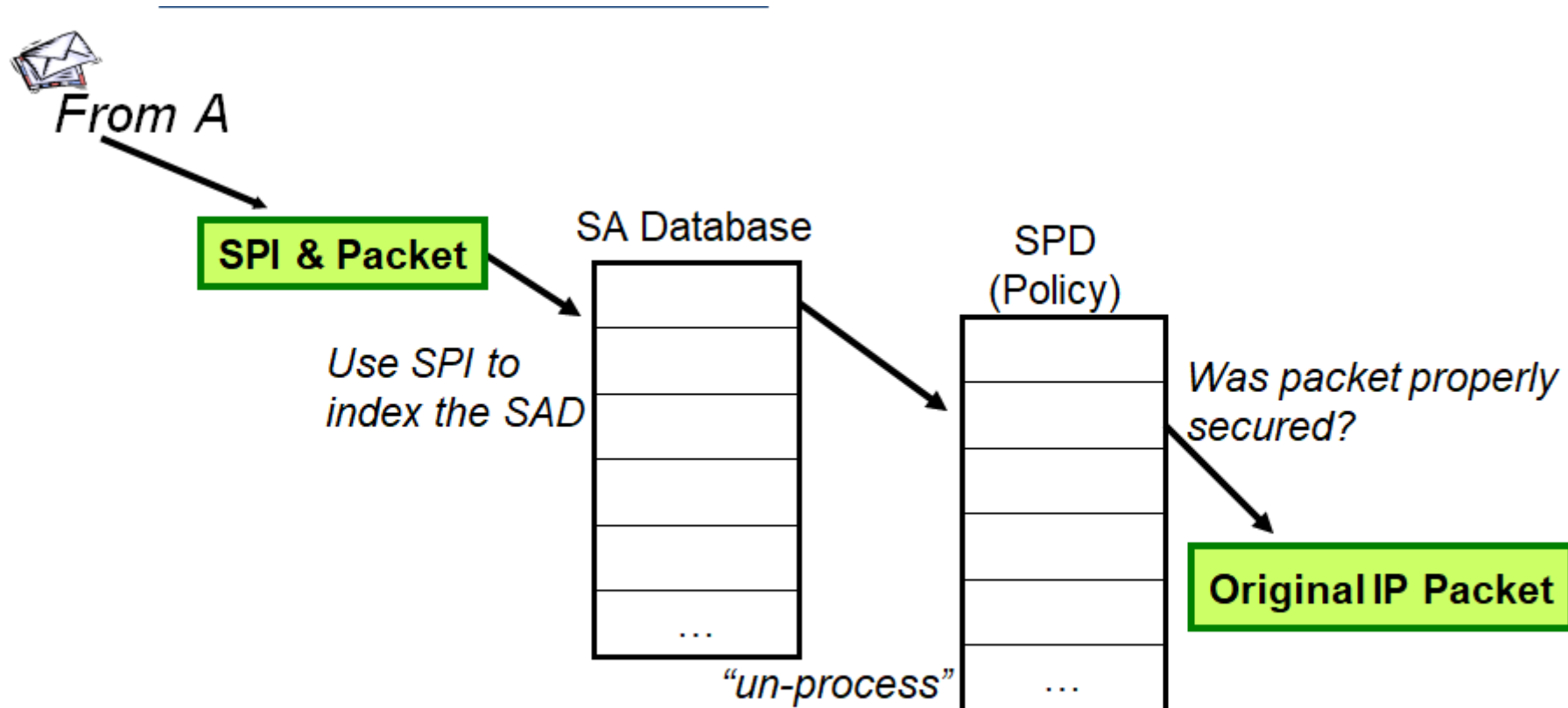
Find Security Association (SA) – If IPsec is required, the system looks up the SA Database to get the encryption keys and parameters, identified by an SPI (Security Parameter Index).

- **Apply IPsec** – The packet is encrypted/authenticated using the SA's settings.
- **Send Packet** – The processed IPsec packet (with SPI) is sent to the destination.

Key Terms:

- **SPD (Security Policy Database)** → Decides *if* and *how* IPsec is applied.
- **SA (Security Association)** → Stores *encryption keys and settings*.
- **SPI (Security Parameter Index)** → Helps the receiver identify the correct SA.
- This ensures secure communication by enforcing policies and applying encryption.

Outbound Process



Work flow of diagram

- **Receive Packet**
 - Gets IPsec packet + **SPI** (identifies security rules).
- **Find SA**
 - Uses **SPI** to look up decryption keys in **SA Database (SAD)**.
- **Decrypt/Verify**
 - Unseals packet using SA's keys.
- **Check Policy (SPD)**
 - Confirms packet was secured correctly.
- **Forward Original Packet**
 - Releases unprotected IP packet for delivery.

- The two diagrams illustrate the **outbound** (sender-side) and **inbound** (receiver-side) IPsec workflows.

Outbound Process (Sender):

The system first checks the **Security Policy Database (SPD)** to determine if the packet requires IPsec protection. If yes, it retrieves the corresponding **Security Association (SA)** from the **SA Database (SAD)** using the **SPI (Security Parameter Index)**. The packet is then encrypted/authenticated and sent with the SPI attached.

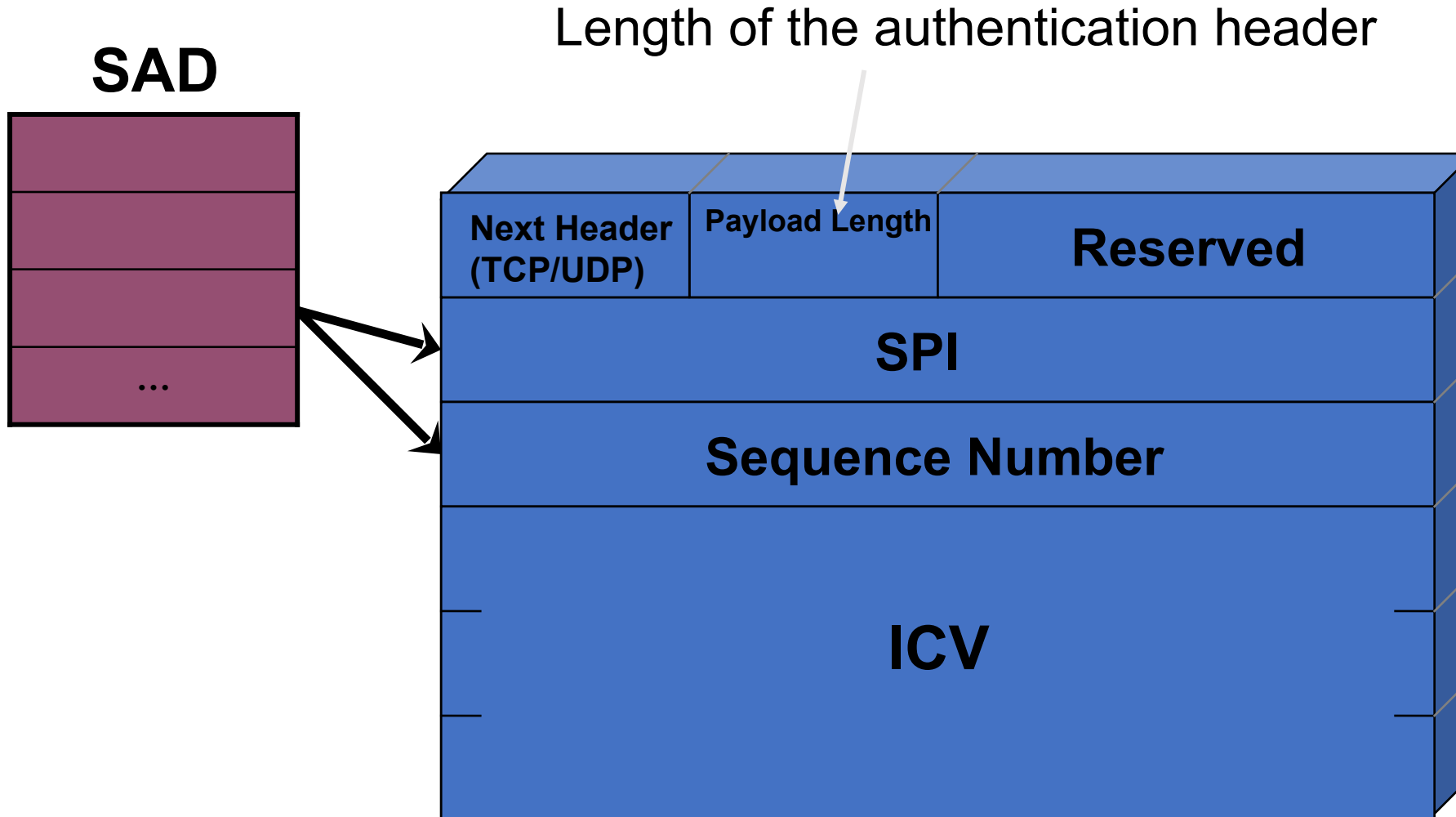
Inbound Process (Receiver):

Upon receiving an IPsec packet, the system uses the **SPI** to locate the correct **SA** in the **SAD** for decryption/verification. After unsealing the packet, the **SPD** verifies if security policies were followed before releasing the original IP packet.

IPsec operates in two main modes: **Tunnel mode** (encrypts entire IP packets, used for VPNs) and **Transport mode** (only encrypts payload, ideal for direct host-to-host communication). At its core, IPsec relies on **Security Associations (SAs)**, which are like contracts defining encryption keys and protocols for secure connections. Each SA is identified by a **Security Parameter Index (SPI)**, acting as a unique ID to locate the right security rules.

Two databases manage these rules: the **Security Policy Database (SPD)** decides *whether* to protect traffic, while the **SA Database (SAD)** stores the *how* (e.g., encryption methods). IPsec uses two key protocols: the **Authentication Header (AH)** for integrity checks, and **Encapsulating Security Payload (ESP)** for encryption (and optionally authentication).

- **AH (Authentication Header)** ensures data integrity and authenticates the sender. It protects the entire packet from tampering and verifies the source IP using a MAC (like HMAC-SHA-1). It also blocks replay attacks with sequence numbers. Unlike ESP, AH doesn't encrypt data—it only guarantees authenticity.



Integrity Check Value (ICV)

- **The Integrity Check Value (ICV)** is a keyed Message Authentication Code (MAC) that ensures data integrity in IPSec. It's calculated over:
- **Static IP header fields** (like source/destination IP) to prevent spoofing,
- **Upper-layer data**, and
- **IPSec headers** (excluding the ICV field itself).
- **Mutable fields** (e.g., TTL, checksum) are skipped to avoid false mismatches. The MAC is often truncated to **96 bits** for efficiency.
- **Purpose:** Guarantees packets are untampered and authentic.

AH: Tunnel and Transport Mode



← authenticated except for mutable fields →



← authenticated except for mutable
fields in the new IP header →



- Original
- Transport Mode
 - Cover most of the original packet
- Tunnel Mode
 - Cover entire original packet

Question?